

# Nithan B S

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## Summary

Data Science and Machine Learning professional specializing in NLP, deep learning architectures, and GenAI workflows. Hands-on experience building, optimizing, containerizing, and deploying functional applications using RAG frameworks, transformers, Docker, and automated APIs.

## Technical Skills

- Data Engineering:** ETL Pipelines, MySQL, Database Design, API Integration, Git Version Control.
- Data Science & ML:** Python (Scikit-learn, Pandas), SQL, Statistical Modelling.
- AI & Deep Learning:** GenAI, LangChain, RAG Pipelines, BERT, NLP, Neural Networks, Hugging Face Transformers.
- Deployment & Tools:** Streamlit, FastAPI, Docker, AWS, Hugging Face Spaces.

## AI & ML Projects

### Legal Intelligence Assistant - Retrieval-Augmented Generation (RAG) Pipeline | [Live Demo](#)

- Developed an end-to-end Retrieval-Augmented Generation (RAG) system for complex legal document question-answering, clause retrieval, and abstractive summarization.
- Implemented vector embeddings and semantic search mechanisms to ensure high precision in document parsing.
- Containerized using Docker and deployed via Git to Hugging Face Spaces, designing custom evaluation workflows incorporating hallucination-aware assessment metrics.

### Financial News Sentiment Prediction using Deep Learning & BERT | [Live Demo](#)

- Architected and trained custom deep learning models (RNN, LSTM, GRU) from scratch, engineering local training loops to establish baseline textual sentiment performance.
- Integrated and evaluated a pre-trained FinBERT model via Hugging Face Transformers to benchmark baseline architectures against state-of-the-art domain-specific embeddings.
- Containerized with Docker and deployed to Hugging Face Spaces, dynamically streaming model weights directly from the Hugging Face Hub at runtime for real-time predictions.

### Marketing Campaign Analysis & Text-to-SQL Agent | [Live Demo](#)

- Executed comprehensive EDA and rigorous data cleaning workflows to identify and resolve underlying data discrepancies.
- Engineered a custom customer segmentation logic based on demographics (income, age, expenditure, and family status) to power an interactive analysis dashboard.
- Architected a SQL backend to handle structured customer data and optimize query performance.
- Engineered an intelligent GenAI layer using a 70B Llama model via Groq API for natural language data retrieval and automated chart generation.

### Real-Time Flight Tracking ETL Pipeline

- Engineered a robust ETL pipeline fetching live data via Aero Databox API to track global flight movements.
- Automated data transformation and cleaning workflows, loading refined datasets into a relational database for real-time querying.
- Implemented version control best practices using Git to ensure code maintainability, scalability, and continuous integration.

### Predictive Modelling for Forest Fire Susceptibility (M.Sc. Dissertation)

- Benchmarked multiple **Machine Learning algorithms** to predict fire risks with high accuracy.
- Processed massive spatiotemporal datasets (NDVI, KBDI) utilizing **Google Earth Engine**.
- Analysed climate teleconnections (ENSO/IOD), showcasing expertise in identifying hidden correlations in noisy data.

## Education

- Master Data Science** | GUVI | 2026
- Integrated M.Sc., Environmental Sciences** | Central University of Rajasthan | 2015-2020

## Professional Development

### Independent Researcher & Competitive Examination Preparation | Chennai | 2020 – 2025

- Pursued full-time preparation for highly competitive national-level examinations, strengthening analytical reasoning, research, problem-solving, and structured decision-making skills.
- Qualified **UGC-NET (2021)** and **PhD Entrance Examinations** in Environmental Sciences.
- Conducted independent study across public policy, economics, governance, environment, and data interpretation.

### Intern | Regional Integrated Multi-Hazard Early Warning System (RIMES) | 2019

- Assisted in hazard modelling, geospatial analysis, and environmental data processing for disaster risk reduction initiatives.
- Supported data-driven assessments for early warning systems and climate resilience planning.
- Collaborated with researchers on analytical workflows involving environmental and hazard datasets.